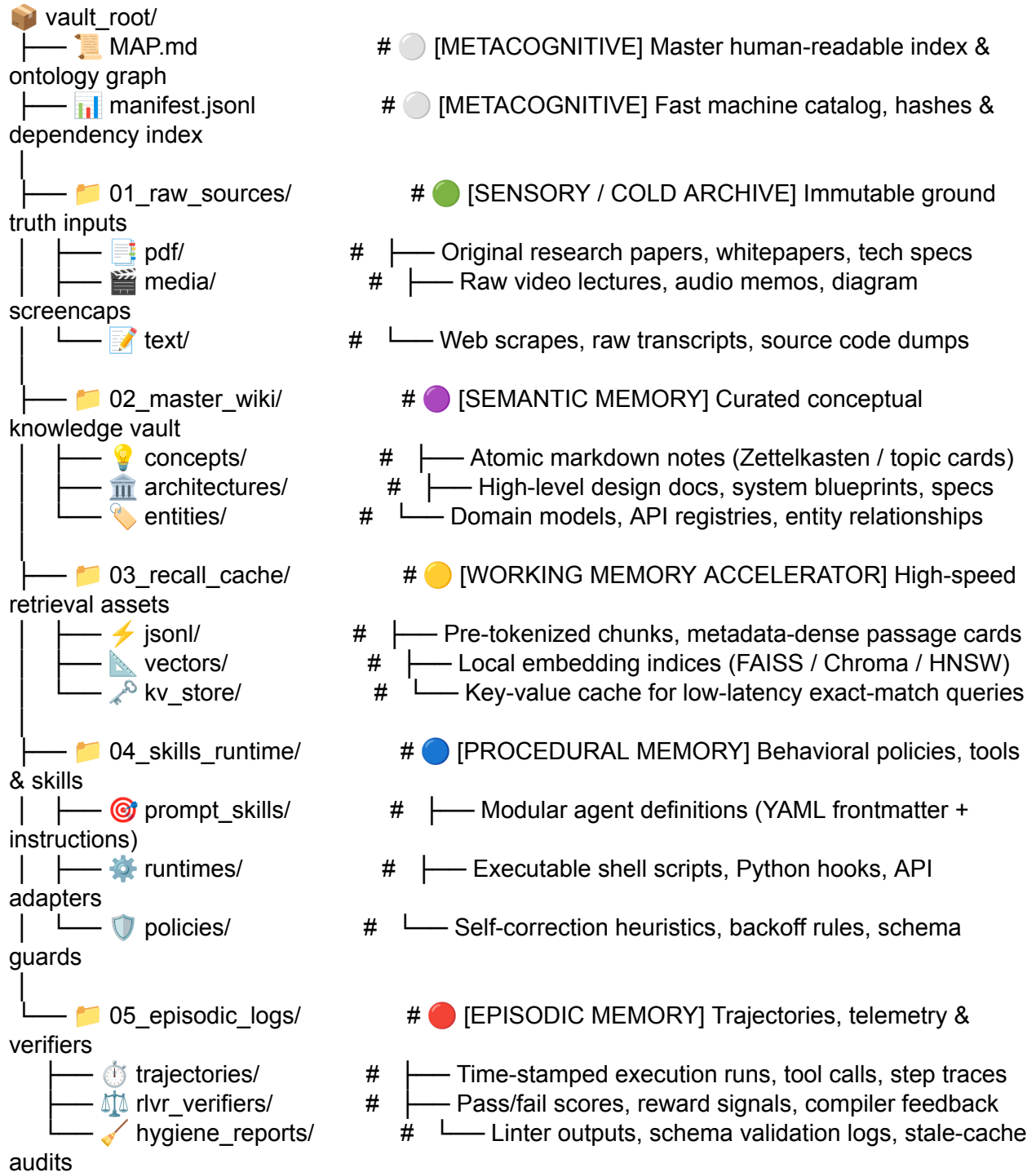


Here is an expanded, visually organized breakdown of the unified architecture. It includes distinct iconography, color-coded functional tags, and detailed descriptions for each tier and subfolder.

Visual Architecture Map

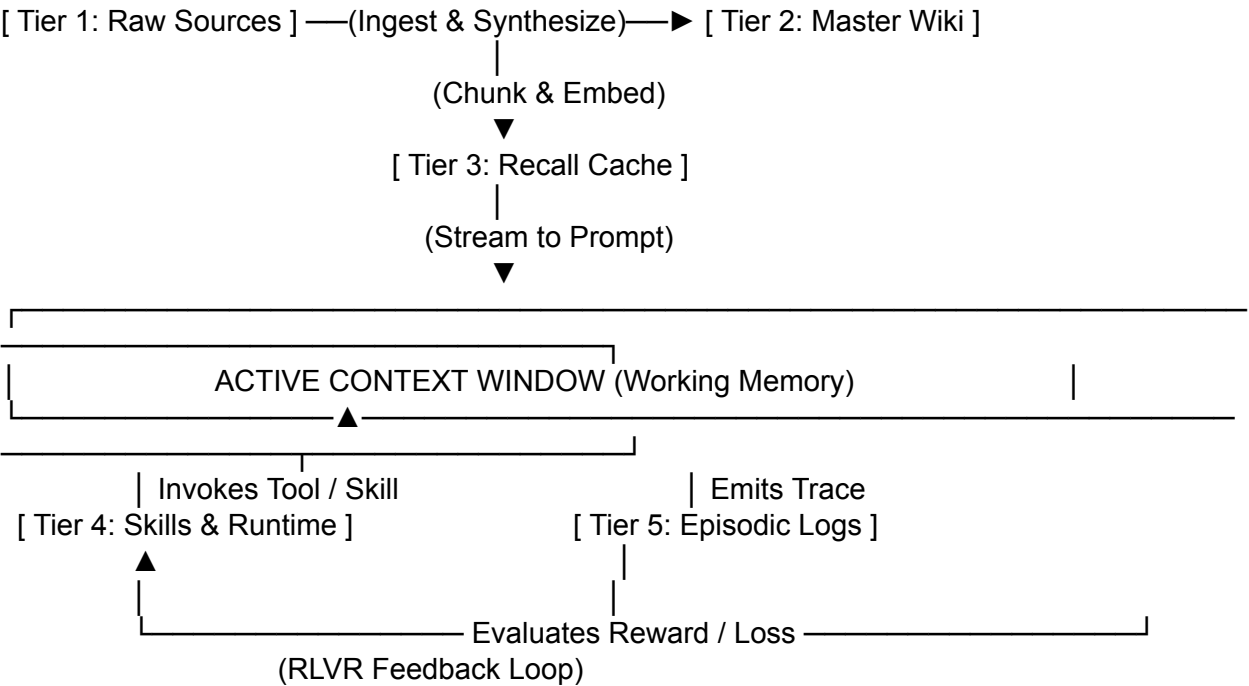


Detailed Tier Breakdown

Tier & Icon	Directory	Cognitive Role	I/O Cadence	Primary Function & Formats
 Metacognitive	/ (Root)	Global Router	Read-Heavy	Contains MAP.md (human navigation) and manifest.jsonl (machine index). Acts as the central directory mapping cross-tier references without loading full file bodies.
 Tier 1	01_raw_sources/	Sensory Buffer	Write-Once	Stores raw, immutable source materials (.pdf, .mp4, .wav, .html). Protected against automated agent modification to guarantee reproducible provenance.
 Tier 2	02_master_wiki/	Semantic Memory	Continuous Sync	Human- and agent-readable Markdown notes (.md) linked via wiki-links. Synthesizes concepts, architectures, and domain schemas extracted from Tier 1.
 Tier 3	03_recall_cache/	Working Memory	Ephemeral / Rebuilt	High-speed machine cache (.jsonl, vector indices). Formatted specifically for sub-second retrieval into the LLM context window during reasoning loops.

Tier & Icon	Directory	Cognitive Role	I/O Cadence	Primary Function & Formats
 Tier 4	04_skills_runtime/	Procedural Memory	Version-Controlled	Houses executable code, prompt skills, and corrective heuristics (.py, .sh, SKILL.md). Encodes <i>how</i> the system works and uses tools.
 Tier 5	05_episodic_logs/	Episodic Memory	Append-Only	Time-series records of agent attempts, trajectory traces, and RLVR reward signals (.log, .jsonl). Provides the training and refinement data for Tier 4.

 Dynamic Data Flow



1. **Extraction:** New inputs hit 01_raw_sources/ and are distilled into atomic notes in 02_master_wiki/.
2. **Indexing:** The wiki is indexed and chunked into 03_recall_cache/ for fast machine retrieval.

3. **Execution:** The agent pulls prompt context from Tier 3, executes tasks using routines from 04_skills_runtime/, and streams every intermediate step into 05_episodic_logs/.
4. **Refinement:** The RLVR verifier grades the episodic logs; passing trajectories become refined skills in Tier 4 or updated architectural notes in Tier 2.